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MILITARY SPECIFICATION

DETONATOR, ELECTRIC, T20E1
PARTS FOR, AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing for one type of electric detonator designated as Detonator, Electric, T20E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-148 - Aluminum Foil
MIL-A-2550 - Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-650 - Explosive: Sampling, Inspection and Testing

FSC: 1336

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MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

* U.S. ARMY ARMAMENT COMMAND

PRODUCT AND PACKING DRAWINGS

8826740 - Detonator, Electric, T20E1, Assembly
8796522 - Marking Diagram and Sealing for Wood Packing Boxes.

INSPECTION EQUIPMENT DRAWINGS

8816368 - Disc, Lead
8871907 - Assembly, Holder
8873526 - Disc, Centering
9200355 - Instruction, Operating
9200356 - Set-Up, Functioning
9200602 - Bushing

(Copies of specifications, standards, drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - TRANSPORATION, PARTS 0-190

The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders shall specify 49 CFR 0-190 (latest revision).

3. REQUIREMENTS

3.1 Materials and Parts. - The materials and parts shall be in accordance with applicable drawings and specifications.

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3.2 Detonators. - The detonators shall comply with all requirements specified in Drawing (Dwg.) 8826740 and with all requirements specified in applicable specifications.

3.3 Moisture Content

3.3.1 Lead Azide. - The moisture content of the lead azide at the loading station at the time of loading the detonators shall not exceed 0.50 percent when tested as specified.

3.3.2 Pentaerythrite Tetranitrate (PETN). - The moisture content of the PETN at the loading station at the time of loading the detonators shall not exceed 0.50 percent when tested as specified.

3.4 Functioning. - The detonators shall comply with the following:

3.4.1 Time. - The detonators shall fire on the first attempt within ten (10) microseconds.

3.4.2 Output. - The detonators shall fire on the first attempt and shall make a 3/16 minimum (min.) diameter perforation in a 0.1345 inch thick lead disc.

3.4.3 Functioning duds.- Detonators that fail to fire on the first attempt are duds. The detonators shall not dud.

3.5 Waterproofness. - The detonators shall comply with the requirements of 3.4.2 and 3.4.3.

3.6 Environmental

3.6.1 Temperature and Humidity. - The detonators shall comply with 3.4.1 after testing as specified in MIL-STD-331, Test 105.1.

3.6.2 Lead circuit resistance, thermal shock, and vibration. - The detonators shall comply with the lead circuit resistance requirement of dwg. 8826740 and with 3.4.1.

3.6.3 Cold Temperature. - The detonator assembly shall comply with 3.4.3.

3.6.4 Hot Temperature. - The detonator assembly shall comply with 3.4.3.

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3.7 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.8 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

3.9 The plug assembly (dwg. 9280965) shall show no evidence of porosity as indicated by air bubbles.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

*4.2 First Article Inspection

4.2.1 Submission. - Prior to the start of regular production the contractor shall submit a first article sample (see 6.1.c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2 and 4.2.3. The first article sample shall consist of 25 complete sets of parts and sub-assemblies (i.e. 25 each of every component and sub-assembly), which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including a certificate of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specifiaction such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. - Assemblies and components will be subjected by the Government to any or all of the examinations and tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection. - If any component fails to comply with the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an inert component or sub-assembly to comply with any of the stated requirements.

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4.2.4 Loaded Detonators. - A sample of three hundred and nine (309) loaded detonators shall be submitted in accordance with the provisions of 4.2.1 for testing as specified in 4.2.4.1.

4.2.4.1 Tests. - The detonators shall be subjected to the following tests:

4.2.4.1.1 Time (see 3.4.1), output (see 3.4.2), duds (see 3.4.3), and waterproofness (see 3.5). - One hundred and eighty-five (185) detonator assemblies shall be selected and tested as specified in 4.3.3.2, Table II.

4.2.4.1.2 Temperature and Humidity (see 3.6.1). - Thirty-seven (37) detonator assemblies shall be selected and tested as specified in 4.4.3.1.

4.2.4.1.3 Lead circuit resistance, thermal shock and vibration (see 3.6.2). - Thierty-seven (37) detonator assemblies shall be selected and tested as specified in 4.4.3.2.

4.2.4.1.4 Cold Temperature Conditioning (see 3.6.3). - Twenty-five (25) detonator assemblies shall be selected and tested as specified in 4.4.3.3.

4.2.4.1.5 Hot Temperature Conditioning (see 3.6.4). - The remaining twenty-five (25) detonator assemblies shall be tested as specified in 4.4.3.4.

4.2.4.2 Rejection. - The loaded detonator first article sample shall be rejected if any of the following test results occur:

4.2.4.2.1 Time, output, duds and waterproofness. - The rejection criteria of 4.3.3.2.2 shall apply.

4.2.4.2.2 Temperature and humidity. - The rejection criteria of 4.3.3.2.2 sub-paragraph "a" shall apply.

4.2.4.2.3 Lead Circuit Resistance, Thermal Shock and Vibration. - The total number of detoantor assemblies that fail to comply with the lead circuit resistance requirement equals or exceeds one, or if, after calculations, the functioning time exceeds the requirement (see 4.3.3.2.2a)

4.2.4.2.4 Cold temperature conditioning. - One or more detonator assemblies fails to fire.

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4.2.4.2.5 Hot temperature conditioning. - One or more detonator assemblies fails to fire.

The Government reserves the right to terminate its inspection upon any failure of an item to comply with any of the stated requirements.

4.3 Inspection Provisions

4.3.1 Lot Formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revision, same specifications and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded detonator assemblies shall contain:

a. Parts of one lot interfix number from one supplier.
b. PETN from not more than one lot.
c. Lead azide of the same type from not more than two lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

4.3.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that

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continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL'as designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup (see dwg. 8826744 covering a detail of dwg. 8826740).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Thickness through bottom, min. .	Gage	01001
102.	Outside diameter, (max.)	Gage	01002
103.	Dent or distortion	Visual	01003
Minor:	AQL 1.00 percent		
201.	Inside diameter, min.	Gage	01004
202.	Evidence of poor workmanship (see 3.7)	Visual	01005

4.3.2.2 Ferrule (see dwg. 8826745 covering a detail of dwg. 8826740)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Length, min.....	Gage	02001
102.	Outside diameter. min.	Gage	02002
103.	Inside diameter, max.	Gage	02003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	02004

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4.3.2.3 Pellet (see dwg. 9280967 covering a detail of dwg. 8826740).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Distance between Electrode holes .	Gage	03001
102.	Diameter of Electrode holes, max..	Gage	03002
103.	Outside diameter	Gage	03003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	03004

4.3.2.4 Plug Assembly (see 9280965 covering a detail of 8826740).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Lead wires not tinned	Visual	04001
102.	Plug cracked	Visual	04002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship other than crack (see 3.7)	Visual	04003

4.3.2.5 Eyelet (see dwg. 9280968 covering a detail of dwg. 8826740).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Outside diameter	Gage	05001
102.	Total length	Gage	05002

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	05003

4.3.2.6 Bridge and plug assembly (see dwg. 9280981 covering a detail of dwg. 8826740)

Categories	Defects	Method Of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Spot charge missing or inadequate	Visual	06001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	06002

4.3.2.7 Assembly, prior to inserting priming charge (see dwg. 8826740).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Weight of base charge, min.	Balance	07001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	07002

4.3.2.8 Assembly, prior to inserting bridge and plug assembly (see dwg. 8826740).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percetn		
101.	Priming charge missing	Visual	08001
102.	Ferrule missing	Visual	08002

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	08003

4.3.2.9 Assembly (see dwg. 8826740)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Lead wires not shorted	Visual	09001
2.	Cup cracked or split	Visual	09002
Major:	AQL 0.65 percent		
101.	Length	Gage	09003
102.	Diameter	Gage	09004
103.	Crimp not full 360 degrees	Visual	09005
104.	Joint seal missing or incomplete.	Visual	09006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	09007

4.3.2.10 Carton, prior to sealing (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Incorrect number of detonators ..	Visual	10001
102.	Detonators in bundle not completely encased in foil	Visual	10002
103.	Packet assembly torn or insecurely closed (when applicable)	Visual	10003
Minor:	AQL 1.00 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	10004
202.	Package marking missing, incorrect or illegible	Visual	10005

4.3.2.11 Carton, sealed (see section 5).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Carton damaged to extent that contents are exposed or liable to become exposed	Visual	11001
Minor:	AQL 1.00 percent		
201.	Marking missing, incorrect or illegible	Visual	11002

4.3.3 Testing. - Unless otherwise specified herein, the sampling plans for the following tests shall be in accordance with MIL-STD-105.

4.3.3.1 Moisture content (see 3.3) - Major defects

TABLE I

Materials	Code Number
Lead Azide	12001
PETN	12002

The contractor shall provide adequate controls to insure compliance with the requirements, and shall test for verification one sample of each material from each eight hours production of detonators. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement, and loading has not begun, that quantity of material represented by the sample shall be rejected. If detonators have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all detontors loaded with the non-conforming material shall be rejected. The tests shall be performed as specified in 4.4.1.1 or 4.4.1.2 as applicable.

4.3.3.2 Functioning and waterproofness (see 3.4 and 3.5).

4.3.3.2.1 Sampling. - One hundred and eighty-five (185) detonator assemblies shall be selected from each lot for test. The sequence of testing and the number of samples to be subjected to each test shall be as specified in Table II.

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TABLE II

Test Sequence	Classification	Code No.	No. of Samples	Test Paragraph
1. Time (see 3.4.1)	Major	09009	35	4.4.2.1
2. Waterproofness (see 3.5) and Output (see 3.4.2)	Major	09010	50	4.4.2.2
3. Duds (see 3.4.3)	Major	09011	*100 less time and output duds	4.4.2.3

*Duds that occur in the time or output phase of testing shall be recorded and replaced from this sample.

4.3.3.2.2 Rejection. - The lot shall be rejected if any of the following test results occur:

a. Time. - The mean functioning time ($\overline{X}$) plus the product of 2.27 and the standard deviation(s) (see 6.6) exceed ten (10) microseconds).

b. Waterproofness and Output. - One or more detonator assemblies fail to produce a perforation of the required size in the lead disc.

c. Duds. - The total number of detonator assemblies that fail to fire on the first attempt in the time, waterproofness and output, and dud tests equals or exceeds two (2).

4.3.3.3 Environmental (see 3.6)

4.3.3.3.1 Sampling. - Eighty-seven (87) detonator assemblies shall be selected from each lot for test. The sequence of test and the number of samples to be subjected to each test shall be as specified in Table III.

TABLE III

Test	Classification	Test Paragraph	No. of Samples	Code No.
1. Lead circuit resistance, thermal shock, and vibration (see 3.6.2)	Major	4.4.3.2	*37 resistance time	09012 09013

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2. Cold temperature (see 3.6.3)	Major	4.4.3.3	25	09014
3. Hot temperature (see 3.6.4)	Major	4.4.3.4	25	09015

*Time calculations shall be based on 35 firings, duds shall be recorded and replacements fired.

4.3.3.3.2 Rejection. - The lot shall be rejected if any of the following functioning results occur:

a. Time after thermal shock and vibration. - The mean functioning time ($\overline{X}$) plus the product of 2.27 and the standard deviation(s) (see 6.6) exceeds ten (10) microseconds.

b. Dud - The total number of detonator assemblies that fail to fire on the first attempt after environmental conditioning as specified in 1, 2, or 3 of Table III equals or exceeds one.

c. Lead Circuit Resistance. - The total number of detonator assemblies that fail to comply with the lead circuit resistance requirement during thermal shock and vibration conditioning equals or exceeds one.

4.3.3.4 Pull Test of Lead Wires (see dwg. 9280965). - Major Defect - Code No. 04006. - The sampling plan for this test shall be in accordance with Code Letter K and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.4.

4.3.3.5 Resistance between lead wires before assembly of wire bridge (see dwg. 9280981). - Major defect - Code No. 13001. - This test shall be performed 100 percent in accordance with the procedures specified in 4.4.5. Any detonator assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.3.3.6 Resistance between lead wires before shorting (see dwg. 8826740) - Major defect. - Code No. 14001. - This test shall be performed 100 percent in accordance with the procedure specified in 4.4.6. Any detonator assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.3.3.7 Insulation resistance between lead wires and cup (see dwg. 8826740) - Major Defect - Code No. 09016. - This test shall be performed 100 percent in accordance with the procedure

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specified in 4.4.7. Any detonator assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.3.3.8 Porosity (see 3.9) - Major Defect - Code No. 04004. - The sampling plan shall be in accordance with Code Letter H and with an AQL of 0.40 percent. The test shall be performed as specified in 4.4.8.

4.3.4 Inspection Equipment. - Inspection equipment required to perform the inspections prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection equipment design and approval.

4.4 Test Methods and Procedures

4.4.1 Mositure Content

4.4.1.1 Lead Azide. - The moisture content of the lead azide shall be determined in accordance with Method 101.5 of MIL-STD-650 except that a two gram (gr.) sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene.

4.4.1.2 PETN. - The moisture content of the PETN shall be determined as specified in 4.4.1.1.

4.4.2 Functioning. - The detonator assemblies shall be temperature conditioned at 70 plus or minus 10 degrees Fahrenheit (f) for two (2) hours min. prior to performing the functioning test.

4.4.2.1 Time. - The detonator assemblies shall be assembled in the test fixture specified in 4.3.4 and fired by applying the energy from a 0.4 max. microfarad capacitor charged to 50 volts max. The elapsed time between application of the energy and detonator functioning shall be measured. Detonators that fail to fire shall be recorded and the replacement tested.

4.4.2.2 Waterproofness and Output. - The detonator assemblies shall be immersed in water at a two (2) to three (3) inch depth for forty-eight (48) hours min. The water temperature shall be maintained at seventy (70) plus or minus 10 degrees F. for the immersion period. At the end of the immersion time, the detonator assemblies shall be removed from the water, assembled into the

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test fixture and fired by applying the same energy specified in 4.4.2.1. The lead disc perforation diameter shall be measured to determine compliance with the requirement. Method of Inspection shall be in accordance with 8816368, 8871907, 8873526, 9200602, 9200355 and 9200356.

4.4.2.3 Duds. - The detonator assembly shall be assembled in the Applicable Test Fixture and fired by applying the energy specified in 4.4.2.1. Observation shall be made for duds. Method of Inspection shall be in accordance with 8816368, 8871907, 8873526, 9200602, 9200355 and 9200356.

4.4.3 Environmental tests

4.4.3.1 Temperature and Humidity. - The detonator assemblies shall be tested in accordance with Test 105.1 of MIL-STD-331. After test thirty-five (35) of the assemblies shall be functioned as specified in 4.4.2.1 and the times recorded.

4.4.3.2 Lead Circuit Resistance, Thermal Shock, and Vibration. - The detonator assemblies shall be tested as follows, in the sequence specified, using equipment in accordance with 4.3.4.

a. Measure resistance of lead circuit in accordance with 4.4.6.
b. Condition for 2 hours min. at minus 65 plus or minus 4 degrees F.
c. Remove from cold chamber and measure lead circuit resistance in accordance with 4.4.6.
d. Condition for 2 hours min. at plus 165 plus or minus 4 degrees F.
e. Remove from hot chamber and measure lead circuit resistance in accordance with 4.4.6.
f. Vibrate for fifteen (15) minutes (min) in a sweep covering a frequency range of 25-500-25 cycles per second parallel to the transverse orthogonal axis with base charge down and a 2 $\pm$ 0.2g peak.
g. Measure resistance of lead circuit in accordance with 4.4.6.
h. The detonator assemblies shall be functioned as specified in 4.4.2.1 and the times recorded.

4.4.3.3 Cold Temperature. - The detonator assemblies shall be conditioned at minus 65 plus or minus 4 degrees F. for two (2) hours min. using equipment specified in 4.3.4. The detonator assemblies maintained at the conditioning temperature shall be functioned as specified in 4.4.2.3.

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4.4.3.4 Hot Temperature. - The detonator assemblies shall be conditioned at plus 165 plus or minus 4 degrees F. for two (2) hours min. using equipment specified in 4.3.4. The detonator assemblies maintained at the conditioning temperature shall be functioned as specified in 4.4.2.3.

4.4.4 Pull Test of Lead Wires. - The plug and wire assembly shall be assembled in the equipment specified in 4.3.4 and the load applied. The wire and plug shall be observed for compliance with the requirement.

4.4.5 Resistance between Lead Wires Before Assembly of Wire Bridge. - The assembly shall be connected to the equipment specified in 4.3.4 and the resistance measured to determine compliance with the requirements.

4.4.6 Resistance Between Lead Wires Before Shorting. - The assembly shall be connected to the equipment specified in 4.3.4 and resistance measured to determine compliance with the requirements.

4.4.7 Insulation Resistance between Lead Wires and Cup. - The detonator assembly shall be connected to the equipment specified in 4.3.4 and the required voltage applied for not less than one nor more than two seconds. The equipment shall be observed to determine compliance with the requirements.

4.4.8 Porosity of plug assembly (see dwg. 9280965). - The plug assembly shall be immersed in an enclosure containing water and a suitable wetting agent. The enclosure shall be evacuated to an absolute pressure equal to 1.3 inches of mercury for one minute, or until air bubbles cease to be given off by the water, whichever is longer. The immersed plug assembly shall be observed for evidence of leakage, indicated by the bubbles emanating from between the pin and plug.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C.

5.1.1.1 Rosette Wrap. - A bundle of 25 detonators shall be made, lead wires out straight, using the lead wires from one detonator to secure the bundle. Each bundle of detonators shall

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be wrapped in a sheet of approximately 8 by 8 by .0015 inch thick aluminum foil complying with MIL-A-148, so that all detonators and lead wires in a bundle are completely encased by the foil. Four bundles of detonators, totaling 100 detonators, shall be packed in a folding carton, set-up box, or comparable container made of material with a minimum board thickness of .03 inch.

5.1.1.2 Pocket Assembly, alternative pack. - The detonators shall be packed in pocket assemblies manufactured in accordance with Figure I (as applicable) made of .0015 inch aluminum foil complying with MIL-A-148. An adhesive seal shall be used to bond the faces of the aluminum foil together to form five pockets. The adhesive used for the sealing of the two faces of the aluminum foil shall be suitable for the application. The strength of the seal shall be sufficient to allow the insertion of the dowel (see Figure I) without separation of the seal. The two faces of any seal shall not be misaligned by more than 1/16 inch when the pouch is formed. Each pocket shall be opened to form a cylinder using a smooth dowel with rounded end and similar to that shown on Figure I. Each pocket may, however, be opened by other suitable methods which will result in providing a satisfactory pocket for insertion of the detonator without injury to the material. One T20E1 Electric Detonator shall be inserted, detonator end down, in each pocket. The lid of the pocket assembly shall be folded over to form a closed pouch. Twenty pocket assemblies or a total of 100 detonators shall be packed in a folding carton, set-up box, or comparable container make of a material with a minimum board thickness of .03 inch.

5.2 Packing

5.2.1 Level C. - Unit packages shall be packed in accordance with the requirements of Code of Federal Regulations CFR 49 0 - 190 requirements for electric blasting caps and in a manner that will insure safe delivery from the manufacturers plant to the first receiving activity.

5.3 Marking. - Each unit package shall be labeled, stamped or printed with the following information:

100 Each - DETONATOR ELECTRIC T20E1

Lot (Mfg. SYMBOL - NO.)

LOADED (Mo/Yr)

DANGEROUS - HANDLE CAREFULLY

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Letters shall be upper case. Size of marking 1/4 inch min. Ink shall be black, waterproof commercial grade.

Labels shall be white paper of 15 pound basic weight min.

Exterior boxes shall be marked in accordance with drawing 8796522. Cleat markings are not applicable when fiber boxes are used. DOT markings shall be:

* (QTY) - ELECTRIC BLASTING CAPS - HANDLE CAREFULLY

* Insert number of detonators in the exterior container.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

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a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. The contractor must include in his submission of designs a listing of any and all Government SIE designs which will be utilized. Partial submission of inspection designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

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6.4 Submission of Ammunition Data Cards. - In addition to the normal distribution of records, one copy of all ammunition data cards shall be forwarded to each of the following:

Commander
Picatinny Arsenal
Dover, New Jersey 07801
ATTN: (1) SARPA-QA-A-M
(2) SARPA-AD-EP-2

6.5 Check test for possible Deterioration. - The following test is designed to prevent detonators that have deteriorated being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require that the check test be performed, the contracting officer should insert in the contract a provision for the performance of the check test. The contracting officer will arrange for the contractor be reimbursed for expense incurred in his performance of the check test for pos sible deterioration. Two hundred (200) detonator assemblies shall be selected from each lot and tested as follows:

6.5.1 Functioning Time and Duds. - Forty (40) detonators shall be subjected to the test specified in 4.4.2.1. Observation shall be made for the functioning time and all duds that may occur shall be recorded. Duds will be replaced by detonators from the 200 selected. If the mean time ($\overline{X}$) plus 2.03 times the standard deviation(s) exceed 10 microseconds, the lot shall be rejected. The standard deviation shall be computed from the formula in 6.6 or its equivalent. Duds shall not enter into the calculation for standard deviation.

6.5.2 Waterproofness, Output and Duds. - Fifty (50) detonators shall be subjected to the test specified in 4.4.2.2. Observation shall be made of the perforation in the disc, and the duds that may occur shall be recorded. Duds shall be replaced by detonators from the 200 selected. If two or more detonators fail to adequately perforate the lead disc, the lot shall be rejected.

6.5.3 Duds. - The remainder of the detonator assemblies selected in 6.5 shall be subjected to the test specified in 4.4.2.3. The lot shall be rejected if the total number of duds accumulated in 6.5.1, 6.5.2 and 6.5.3 equals or exceeds four (4).

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6.6 Calculation of sample standard deviation.-The following formula or equivalent shall be used in calculating the sample standard deviation:

$$s = \sqrt{\frac{\Sigma(X - \overline{X})^2}{n - 1}}$$

where:

s = sample standard deviation.
$\underline{x}$ = numerical value of individual observations.
$\overline{X}$ = average value of the sample
n = sample size

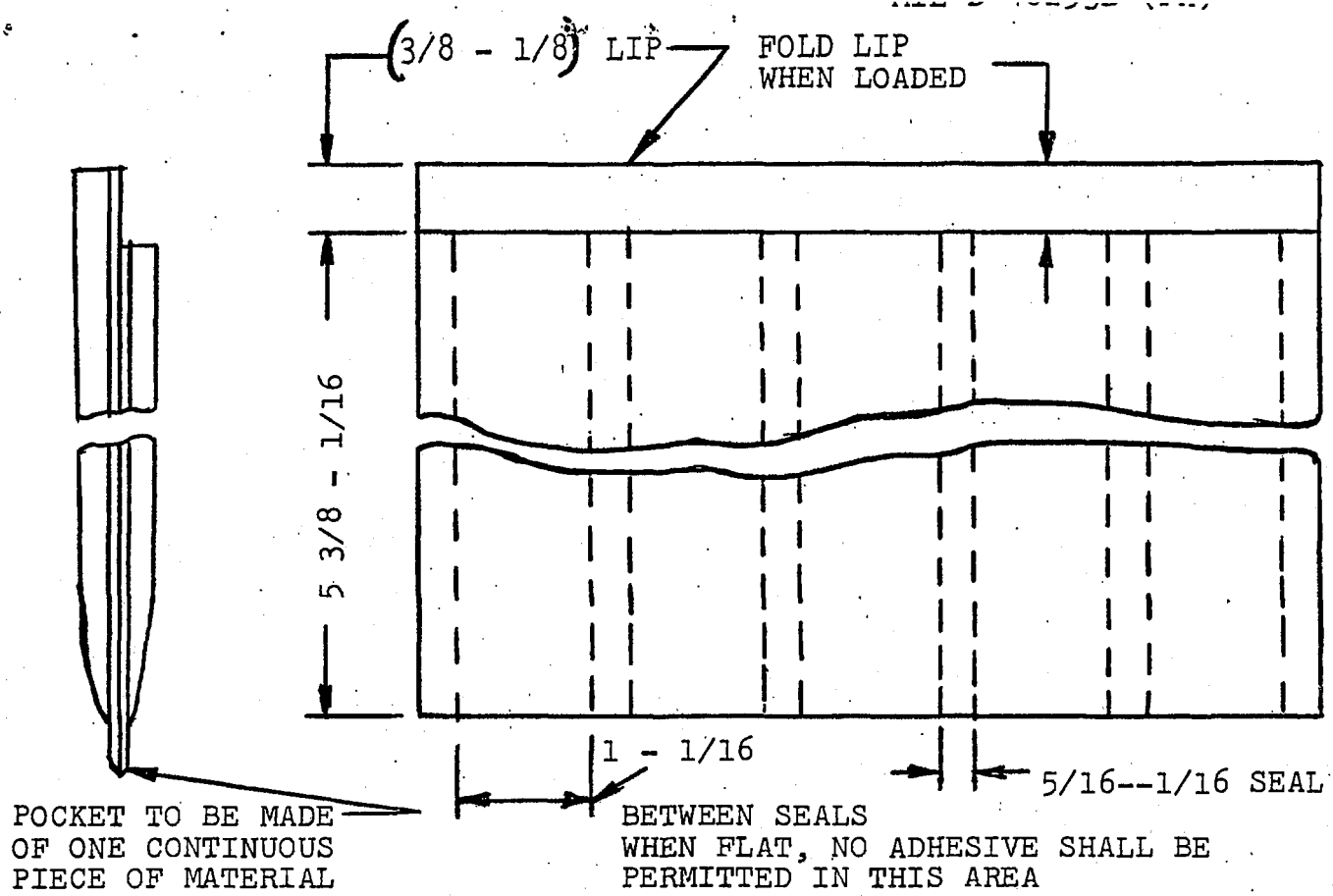
*6.7 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army-PA

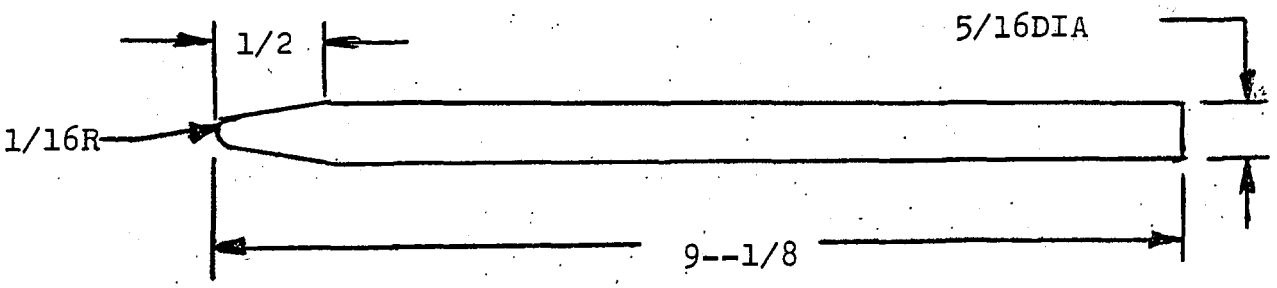
Preparing Activity:
Army-PA

Project Number: 1336-A-219

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POCKET ASSEMBLY



DOWEL SHALL BE SMOOTH WITH NO SHARP CORNERS

DOWEL

FIGURE I

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